

# Dr. Frederik Bous

## Résumé

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### Research interest

AI for music, voice, and sound: Controllable and interpretable generative models, self-supervised representation learning, music information retrieval, neural audio codecs and processing.

### Experience

**Supertone Inc., Korea** 2024 – today

#### Research Scientist

- Develop, train, and package Supertone's voice backbone NANSY
  - Improve controllability through improved pitch encoder
  - Optimize model architecture for better synthesis quality and faster inference
- Experiment with LLM-based TTS and neural audio codecs
- Develop and maintain voice editing tool built on Supertone's voice technology
- Publish and present research at international conferences

**IRCAM, France**

2018 – 2024

#### Post-doc, Doctoral researcher, Intern

- Research on deep learning for voice
  - Topics: Neural vocoding and representation, self-supervised learning, voice attribute transformation.
- Technical support for film, music, and art: bespoke voice transformation for artists.
- Application development: research results into voice editing and live transformation apps.

### Education

**Doctor of Philosophy (PhD)** (Information and Communication) *Sorbonne Université, France* 2023

**Master of Science (MSc)** (Computational Engineering) *TU Darmstadt, Germany* 2019

**Bachelor of Science (BSc)** (Electrical Engineering) *TU Darmstadt, Germany* 2014

### Skills

**Research** Experimental design and evaluation · scientific writing; presented at ICASSP, Interspeech, ISMIR

**Machine learning** Generative modeling · self-supervised learning · neural codecs and vocoders

**Domain knowledge** Signal processing · acoustics · voice science · music theory

**Engineering** Python (PyTorch, NumPy) · application development · model deployment

**Languages** German (native) · English (C2) · French (B2) · Polish (B1) · Chinese (A2) · Korean (basic)

**Music** Classically trained singer · composition · live coding (Tidal Cycles) · piano

### Other

**Full PhD scholarship** *EDITE de Paris (ED130)* 2019 – 2022

**Full study scholarship** *Studienstiftung des deutschen Volkes (SdV)* 2015 – 2019

**Additional credits** (Maths & Music) *TU Darmstadt, Germany* 2015 – 2018

**Internship** *Siemens AG, Germany* 2015

**Study exchange** *Tongji University, Shanghai, China* 2014 – 2015

**Teaching assistant** (Stochastic signals and systems) *TU Darmstadt, Germany* 2014

**Deutschlandstipendium** *TU Darmstadt, Inro Elektrotechnik, Siemens* 2013 – 2014